

SHREYA REDDY LETHAKULA

AI/ML/Data Engineer

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PROFESSIONAL SUMMARY

- Data Scientist / ML Engineer with 2 years of professional experience building and deploying Large Language Model (LLM) applications, Retrieval-Augmented Generation (RAG) pipelines, and production ML systems across AWS and GCP, with hands-on Azure experience for GPU-accelerated model training.
- Skilled in end-to-end ML pipeline development with Python, PyTorch, and TensorFlow, LLM fine-tuning (LoRA/PEFT), and MLOps practices including CI/CD, MLflow, and Docker/Kubernetes.
- Experienced in designing and orchestrating multi-agent GenAI systems with LangChain and LangGraph, integrating vector databases such as Pinecone for semantic search, and fine-tuning domain-specific LLMs for enterprise use cases.
- Skilled in building predictive analytics and NLP solutions — including cybersecurity risk prediction, patient risk stratification, readmission forecasting, and clinical text de-identification — using Random Forest, XGBoost, and Neural Networks, with SHAP/LIME for model explainability.
- Comfortable working full-stack, from data pipelines and model development through React.js front-end interfaces and REST API integration, with hands-on experience automating and monitoring workflows using Airflow, Kubeflow, Prometheus, and CI/CD tooling.
- Strong communicator who translates experimental results and technical findings into measurable, actionable business impact for cross-functional stakeholders.

TECHNICAL SKILLS

Languages: Python, R, SQL, Java, C#, JavaScript

AI/ML & LLMs: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, Hugging Face Transformers, LangChain, LangGraph, RAG pipelines, Prompt Engineering, LoRA/PEFT fine-tuning, GPT-4, LLaMA, Falcon, Claude

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Tableau, Power BI

Cloud & MLOps: AWS (EC2, S3, SageMaker, Cognito, ECS Fargate), Azure (VMs, CUDA), GCP (GKE, BigQuery, Vertex AI, Cloud Functions), MLflow, Kubeflow, Airflow, Prometheus, Docker, Kubernetes, CI/CD (Jenkins, GitHub Actions)

Other: Vector Databases (Pinecone), Apache Spark, Kafka, Snowflake, A/B Testing, NLP/NER, SHAP/LIME

PROFESSIONAL EXPERIENCE

GenAI/ML Data Engineer

Jan 2025 –

Present

Cortracker Inc, Irving, TX

- Designed and deployed Retrieval-Augmented Generation (RAG) pipelines integrating GPT-4, Claude, LLaMA, and Falcon via LangChain, with Hugging Face Transformers for embeddings and Pinecone vector search to power clinical knowledge retrieval and automated NER for diagnoses, medications, and procedures from EHR data.
- Built a multi-agent GenAI system with LangGraph to orchestrate patient risk stratification, readmission prediction, and disease progression forecasting workflows on top of Random Forest, XGBoost, and Keras/TensorFlow Neural Network models, using SHAP/LIME to explain model outputs for clinical stakeholders.
- Fine-tuned domain-specific LLMs using LoRA/PEFT techniques on Azure CUDA VMs for clinical note summarization and HIPAA-compliant PHI de-identification, improving accuracy while reducing manual documentation time.
- Deployed LLM and ML models to production on AWS SageMaker and ECS Fargate with Cognito-based authentication, and on GCP Vertex AI/GKE with BigQuery and Cloud Functions for scalable fine-tuning, data processing, and inference; monitored deployments with Prometheus for real-time performance and latency tracking.
- Automated ETL/streaming pipelines with Kafka, Spark, and Databricks for real-time multi-source healthcare data, and used Snowflake for centralized data warehousing; integrated FHIR standards for interoperability.
- Built Power BI dashboards for patient risk scores and treatment outcomes, and conducted A/B testing of AI-powered patient engagement tools, measuring improvements in communication and appointment adherence.

PROJECTS (ACADEMIC — MASTER'S PROGRAM)

BookNest — *Library Management Analytics Platform*

- Worked as a full-stack developer building both the front end and back end: designed responsive UI components and pages in React.js, and built back-end services in Node.js with MongoDB, deployed on AWS (EC2, S3).
- Built database queries and aggregation pipelines in MongoDB to automate data processing, optimize book retrieval patterns, and improve performance.
- Ran A/B tests on search and recommendation feature variants to measure impact on user engagement and accessibility, using results to guide the final front-end design.
- Developed responsive dashboards and reporting functionality to support data-driven decisions on inventory and user activity, with role-based authentication for data security.

Investire — *Startup-Investor Intelligence Platform*

- Built the platform end to end as a full-stack developer, developing React.js front-end interfaces and back-end REST APIs in Node.js with MongoDB, hosted on AWS.
- Ran A/B tests on investor-facing navigation and dashboard layouts, analyzing user engagement and interaction data to improve decision-making efficiency.
- Designed analytical dashboards and reporting features on the front end to monitor funding trends, user activity, and platform performance metrics.
- Implemented back-end database optimization on MongoDB and AWS-based API integration strategies to improve data retrieval speed and scalability for real-time insights.

Prediction of Cyber Hacking Breaches — *ML for Cybersecurity Risk*

- Developed and trained machine learning models in Python (Scikit-learn) using statistical analysis techniques to predict potential cybersecurity breaches.
- Applied classification algorithms, feature engineering, and multivariate analysis to identify attack patterns and high-risk behaviors.
- Performed data preprocessing, anomaly detection, and model evaluation to improve predictive accuracy and reliability.
- Built a React.js front-end dashboard to visualize risk insights and deliver actionable recommendations, supporting proactive threat mitigation.

EDUCATION

Master of Science, Computer Science

2022 – 2024

University at Albany, SUNY — Albany, NY

CERTIFICATIONS

- Generative AI with Large Language Models — DeepLearning.AI & AWS (Coursera)
- MLOps | Machine Learning Operations (Udemy)
- LangChain & LLM App Development: ChatGPT, OpenAI, RAG, Vector DBs (Udemy)
- Foundations of AI and Machine Learning Microsoft (Coursera)
- AI and Machine Learning Algorithms and Techniques Microsoft (Coursera)
- Building Intelligent Troubleshooting Agents Microsoft (Coursera)
- Microsoft Azure for AI and Machine Learning Microsoft (Coursera)
- Advanced AI and Machine Learning Techniques and Capstone Microsoft (Coursera)